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Original Article

Gaps in Glycaemic Monitoring and Underuse of Cardioprotective Glucose-lowering Agents: Challenges of inpatient diabetes care identified by EFIM's Diabesity Day Survey 2024

Ann-Kristin Porth [a,*], Anete Palma [b], Burcu Sallarel [c], Dimitrios Patoulias [d], Dirk Müller-Wieland [e], Dror Dicker [f], Ieva Ruža [g], Ilya Davidenko [h], Jan Škrha jr. [i], Julia Brandts [e], Ricardo Gomez-Huelgas [j], Shiran Gruber [f], Sindija Smirnova [b], Theocharis Koufakis [d], Yusuf Ziya Sener [k], Zuzana Kršáková [i], Alexandra Kautzky-Willer [a], on behalf of the EFIM Diabetes and Obesity Working Group [1]

[a] Medical University of Vienna, Department of Medicine III, Division of Endocrinology and Metabolism, Vienna, Austria
[b] Department of Internal Medicine, Pauls Stradins Clinical University Hospital, Riga, Latvia
[c] Department of Internal Medicine, Kanuni Sultan Suleyman Training and Research Hospital, Istanbul, Turkey
[d] Second Propaedeutic Department of Internal Medicine, Hippokration General Hospital, Aristotle University of Thessaloniki, Thessaloniki, Greece
[e] Department of Medicine I, University Hospital RWTH Aachen, Aachen, Germany
[f] Rabin Medical Center, Hasharon Hospital, Department of Internal Medicine D and Obesity Clinic, Petah Tikva, Israel
[g] Department of Endocrinology, Riga East Clinical University Hospital, Riga, Latvia
[h] Departement of Internal Medicine 3, Rostov State Medical University, Rostov-on-Don, Russia
[i] 3rd Dept. of Internal Medicine, First Faculty of Medicine and General University Hospital in Prague, Czech Republic
[j] Internal Medicine Department, Regional University Hospital of Málaga, Biomedical Research Institute of Málaga (IBIMA), University of Málaga (UMA), Málaga, Spain
[k] Department of Cardiology, Thoraxcenter, Erasmus MC, Rotterdam, The Netherlands

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ABSTRACT

Background: People with diabetes and obesity have an increased risk of hospitalisation and in-hospital complications. Rising prevalence of diabetes and obesity, including their co-occurrence, "diabesity", make guideline-compliant treatment increasingly important. However, evidence on specific in-hospital diabesity treatment is limited. We aimed to characterise inpatients with diabetes/diabesity and evaluate their in-hospital care to identify challenges in inpatient diabetes/diabesity management.

Methods: Cross-sectional data was collected from patient records and ward charts. We analysed differences between inpatients with and without diabesity and explored in-hospital diabetes treatment. We further compared inpatients with type 2 diabetes who started insulin treatment or received cardioprotective glucose-lowering agents to those who did not and used logistic regression to identify predictors of insulin initiation and use of cardioprotective agents.

Results: We included 207 people with diabetes from eight European hospitals, 50 % with diabesity. Most inpatients had a HbA1c >6.5 % (48 mmol/mol). Among inpatients with type 2 diabetes, one third did not have a recent HbA1c reading, blood glucose levels were monitored <3 times daily, and only 40 % of those with cardiovascular/renal disease received cardioprotective therapies. HbA1c, creatinine and at-home medications predicted insulin initiation, while admission cause and BMI predicted use of cardioprotective agents.

* Corresponding author.
E-mail address: ann-kristin.porth@meduniwien.ac.at (A.-K. Porth).

[1] Representatives of the EFIM Diabetes and Obesity Working Group: Prof. Rachid Malek (Algerian Society of Internal Medicine), Dr. Claudia Ress (Austrian Society of Internal Medicine), Dr. Savvas Ioannou (Cyprus Associaton of Internal Medicine), Dr. Jan Škrha Jr. (Czech Republic Society of Internal Medicine), Dr. Dirk Müller-Wieland (German Society of Internal Medicine), Dr. Dimitrios Patoulias (Hellenic Society of Internal Medicine), Prof. Dror Dicker (Israeli Society of Internal Medicine), Dr. Stephen Malnick (Israeli Society of Internal Medicine), Prof. Caterina Conte (Italian Society of Internal Medicine), Dr. Ieva Ruza (Latvian Society of Internal Medicine), Prof. Diogo Cruz (Portuguese Society of Internal Medicine), Dr. Ilya Davidenko (Russian Scientific Society of Internal Medicine), Prof. Peter Jackuliak (Slovakian Society of Internal Medicine), Prof. Francisco Javier Carrasco Sánchez (Spanish Society of Internal Medicine), Dr. Javier Ena Muñoz (Spanish Society of Internal Medicine), Dr. Nauman Jadoon (Royal College of Physicians of Edinburgh), Prof. Alexandra Kautzky-Willer (Austrian Society of Internal Medicine)

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Conclusions: We observed an underuse of cardioprotective glucose-lowering therapies, low rates of glycaemic monitoring, and low availability of HbA1c readings, concluding that clinical guidelines are not sufficiently implemented in everyday practice in Europe. Based on this, we advocate for better staff training, involvement of diabetologists, and raising awareness of the benefits of cardioprotective agents in the hospital.

1. Background

People with diabetes and people with obesity have an increased risk of hospitalisation, tend to have longer hospital stays, and have higher mortality when hospitalised (1–3). Diabetes management during inpatient care is challenging for healthcare staff. Maintaining blood glucose levels within the recommended target range (100–180mg/dL / 5.6–10.0mmol/L) is crucial for treatment outcomes (4). Hyperglycaemia during hospitalisation is associated with increased rates of complications and mortality (2) which is why regular monitoring of blood glucose levels with point-of-care (POC) capillary testing before meals and at night-time is recommended (4). With the rising prevalence of diabetes and obesity ensuring guideline-compliant treatment before and during hospitalisation becomes increasingly important.

The simultaneous occurrence of diabetes and obesity, i.e., "diabesity", compounds the health risk of either condition alone (5). Both are part of the cardiovascular-kidney-metabolic (CKM) syndrome, which is further characterised by chronic kidney disease (CKD) and (risk of) cardiovascular disease (CVD) (6). The fact that diabesity increases the risk for comorbidities affects treatment recommendations, which include the use of medications with a weight neutral or weight-loss effect, such as metformin, dipeptidyl peptidase 4 inhibitors (DPP-4i), sodium-glucose transporter 2 inhibitors (SGLT-2i), glucagon-like peptide 1 receptor agonists (GLP-1ra), and dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 receptor agonists (5). SGLT-2i and GLP-1ra are particularly recommended for people with diabetes and CVD, severe obesity, and/or uncontrolled hyperglycaemia due to their cardioprotective effects (6). In contrast, insulin has long been the preferred treatment for most non-critically ill inpatients with diabetes experiencing hyperglycaemia during hospitalisation (7,8). However, based on growing evidence on safety, efficacy, and benefits of non-insulin agents in the hospital setting, specifically DPP-4i, SGLT-2i, and GLP-1ra, most recent diabetes guidelines recommend their continued use or initiation during hospitalisation, in particular for patients without severe hyperglycaemia and/or severe acute illness (4,9,10).

According to guidelines, in-hospital diabetes treatment should be individualised (4). Whether and how diabesity may currently be factored into in-hospital treatment adaptations is unclear. Despite existing clinical guidelines for inpatient diabetes/hyperglycaemia management (4,11,12) and recommendations for the general management of diabesity (5) and CKM (6), evidence on specific in-hospital treatment strategies for people with diabesity and/or CKM is limited. Exploring how these at-risk groups differ (e.g., regarding diabetes treatment, admission causes, or blood glucose management) may help close this gap. Prior cross-sectional studies conducted in Spanish hospitals revealed substantial discrepancies between diabetes guidelines and actual inpatient diabetes management (13,14). We aimed to expand and update these findings with information about the status of current inpatient diabetes management in Europe. Our objective was to characterise inpatients with diabetes, particularly those with diabesity, and to evaluate diabetes management, including pharmacological treatment, blood glucose monitoring, and glycemia, as well as to test the feasibility of coordinated data collection within the EFIM Network. The aim is to periodically repeat the data collection and highlight key challenges in inpatient diabetes management.

2. Methods

This cross-sectional study aimed to characterise people with diabetes/hyperglycaemia hospitalised on World Diabetes Day 2024. It was conceptualised by the European Federation of Internal Medicine's (EFIM) Diabetes and Obesity Working Group. Eight affiliated medical centres participated. Ethical approval was received at the Medical University of Vienna (EK 1828/2024), RWTH Aachen University (EK 24–369), Kanuni Sultan Suleyman Training and Research Hospital (Decision No. 236), Riga East Clinical University Hospital (AP/08–08/24/47), Rostov State Medical University (#21/24), and Rabin Medical Centre (0477–24 – RMC). At the General University Hospital in Prague and Hippokration Hospital of Thessaloniki (18/2–10–2024), the study was categorised a public health surveillance activity not requiring formal ethical approval. The study was registered at clinicaltrials.gov (NCT06680674).

2.1. Participants and data collection

We used convenience/census sampling including all people with known and newly diagnosed diabetes (random glucose >200mg/dL) or stress hyperglycaemia (random plasma glucose >140mg/dL on admission) that were hospitalised in internal medicine wards of participating medical centres on November 14th, 2024.

Data collection was guided by the "Diabesity Day Survey", inspired by a similar survey conducted in Spanish hospitals (13,14), and included information on hospital characteristics, patient characteristics, hospital admission, diabetes and obesity treatment, and in-hospital blood glucose management (Supplemental Table 1). The information was gathered from patient records and ward charts, including everything from the day of admission to November 14th, 2024. Anonymised data was entered into a joint research platform hosted by the Medical University of Vienna. A hospital ID was assigned to each medical centre and recorded with the participant data to allow for contacting the responsible researcher in case of ambiguities within the data.

2.2. Data analysis

The data was exported for analysis on March 25th, 2025. Prior to the export, it was manually checked for plausibility by the first author. In case of implausible/ambiguous data, the medical centre providing the data was de-identified and the principal researcher contacted to check and confirm their data.

Diabetes complications and comorbidities were identified from the Charlson comorbidity index (13,15): Myocardial infarction (MI), peripheral artery disease (PAD), cerebrovascular disease (CD), diabetes with end-organ damage, and renal disease were used to establish the presence and number of diabetes complications; with MI, PAD, and CD summarized as CVD. The remaining options[2] were used to establish the presence and number of comorbidities. Moreover, we categorized a person as frail if they scored ≥5 on the Clinical Frailty Scale (16).

First, we characterised all inpatients with diabetes/hyperglycaemia included in this study using descriptive statistics, including central tendencies (mean and standard deviation or median and interquartile

[2] Congestive heart failure, dementia, chronic obstructive pulmonary disease (COPD), connective tissue disease, peptic ulcer disease, liver disease, hemiplegia, acquired immunodeficiency syndrome (AIDS), and cancer

range) and frequency distribution (N and %). We used pairwise deletion (available-case analysis) to deal with missing data. Additionally, we separately analysed inpatients with type 1 and type 2 diabetes. We then focused on people with type 2 diabetes, comparing inpatients with normal BMI (18.5–24.9 kg/m^2), overweight (BMI=25–29.9), and obesity (BMI>30kg/m^2). We used independent *t*-tests, Wilcoxon-Mann-Whitney tests, and Chi-square tests to analyse the differences between inpatients with diabesity compared to those with type 2 diabetes and normal weight.

Second, we explored in-hospital type 2 diabetes management, including glycaemic monitoring and pharmacological treatment. Inpatients with known type 2 diabetes only were included in these analyses. We describe in-hospital medication use, including the initiation or discontinuation of medications during hospitalisation and the number of diabetes medications used during hospital stay compared to before. We place particular focus on the initiation of insulin therapy and the use of cardioprotective glucose-lowering therapies, i.e., SGLT-2i and GLP-1ra. Again, we used independent *t*-tests, Wilcoxon-Mann-Whitney tests, and Chi-square tests to analyse the differences between inpatients who started insulin treatment or who received cardioprotective glucose-lowering therapies during hospitalisation and those who did not. Moreover, we used multivariable binomial logistic regression to identify predictors of insulin treatment initiation (yes/no) and multivariable multinomial logistic regression to identify predictors of the in-hospital use of SGLT-2i and GLP-1ra (SGLT-2i only/ GLP-1ra only/ both/ neither). Before fitting the models, we imputed missing data using multivariate imputation by chained equations (mice). We fitted both models twice, once with the complete dataset and once after removing data from centres with a sample size $N < 18$, to test for potential centre effects. For the binomial regression, we employed backward elimination, starting with a selection of potential predictor variables based on our observations and expert knowledge. The variable with the highest p-value was sequentially removed until all variables remaining were statistically significant. For all models, we report odds ratios, 95 % confidence intervals, p-values, and the AIC. We employed a significance level of $\alpha \leq 0.05$ for all analyses and used the Bonferroni-Holm method to correct for multiple testing.

3. Results

3.1. Characteristics of inpatients with diabetes

Our analysis included 207 people with diabetes from eight hospitals, corresponding to about 30 % of all inpatients on participating wards (N = 704). Most common causes of admission were CVD (33.3 %), respiratory diseases (9.7 %), and infections (9.2 %). Most participants (85.5 %) had type 2 diabetes and ten participants each had type 1 diabetes, another type of diabetes, or stress hyperglycaemia. Information about the entire cohort and for people with type 1 and type 2 diabetes separately are summarised in Table 1. While almost 50 % of people with type 2 diabetes had diabesity (BMI ≥30kg/m^2), none of the people with type 1 diabetes had obesity. In contrast, one third of them were underweight (BMI <18.5kg/m^2).

We observed subtle differences between inpatients with type 2 diabetes and normal weight, overweight or obesity (Table 2). Inpatients with diabesity (N = 76) were more often admitted to hospital due to CVD (43.4 % vs 18.4 %; p=.003), treated with metformin, both at home (60.5 % vs 40.8 %; p=.049) and in hospital (47.4 % vs 22.5 %; p=.001), and more likely to receive obesity management medication (14.5 % vs 2.1 %; p=.031) compared to people with type 2 diabetes and normal weight (N = 49). There was a tendency towards more people with diabesity being female, living with frailty or liver disease, and receiving less POC blood glucose measurements. People with overweight appeared more likely to be admitted to hospital for diabetes-related causes and experience in-hospital hypoglycaemia. After correction for multiple testing none of these differences remained statistically significant

Table 1
Cohort characteristics.

Variable	All N = 207	Type 2 diabetes N = 177	Type 1 diabetes[i] N = 10
Age (Years)	69.0 (14.5)	71.3 (12.2)	38.9 (15.3)
Sex:			
Female	94 (45.4 %)	81 (45.8 %)	6 (60.0 %)
Male	113 (54.6 %)	96 (54.2 %)	4 (40.0 %)
Diabetes duration (Years)	12.5 (10.1)	12.7 (9.7)	9.4 (7.5)
BMI (kg/m^2)	29.1 (6.7)	29.8 (6.6)	22.0 (5.0)
<18.5	6 (3.1 %)	2 (1.2 %)	3 (33.3 %)
18.5 – 24.9	60 (31.1 %)	49 (29.7 %)	3 (33.3 %)
25 – 29.9	46 (23.8 %)	38 (23.0 %)	3 (33.3 %)
≥30	81 (42.0 %)	76 (46.1 %)	0
HbA1c (%)	7.4 (2.6)	7.2 (2.3)	10.9 (3.9)
>6.5 % (>48mmol/mol)	72 (52.0 %)	61 (51.3 %)	8 (88.9 %)
Creatinine (mg/dL)	1.38 (1.09)	1.45 (1.14)	0.89 (0.20)
Admission cause:			
Diabetes-related[ii]	31 (15.0 %)	23 (13.0 %)	6 (60.0 %)
Not diabetes-related[iii]	176 (85.0 %)	154 (87.0 %)	4 (40.0 %)
Hosptalization duration[iv] **(Days)**	9.6 (10.9)	9.7 (11.0)	5.5 (5.2)
Smoking status:			
Current	38 (19.7 %)	29 (17.7 %)	3 (30.0 %)
Former	67 (34.7 %)	59 (36.0 %)	2 (20.0 %)
Never	88 (45.6 %)	76 (46.3 %)	5 (50.0 %)
Frail[v]	130 (67.7 %)	115 (69.7 %)	2 (28.6 %)
Diabetes complications[vi]	136 (65.7 %)	123 (69.5 %)	3 (30.0 %)
Comorbidities[vii]	143 (69.1 %)	122 (68.9 %)	3 (30.0 %)
Charlson comorbidity index (CCI)	7.0 (4.0; 9.0)	7.0 (5.0; 9.0)	1.5 (0.3; 2.0)
Use of diabetes technology[viii]**:**	5 (2.5 %)	1 (0.6 %)	3 (33.3 %)
At home	5 (2.5 %)	1 (0.6 %)	3 (37.5 %)
During hospital stay			
Obesity treatment[ix]	18 (8.9 %)	17 (9.8 %)	1 (11.1 %)
No. of point-of-care blood glucose measurements/ day	2.8 (1.8)	2.6 (1.4)	6.2 (3.7)
Blood glucose (mg/dL)[x]	152.6 (53.6)	154.6 (54.6)	152.7 (57.0)
Hypoglycaemia (<70mg/dL)	15 (7.9 %)	9 (5.6 %)	5 (50.0 %)
Severe hypoglycaemia (<40mg/ dL)	6 (3.1 %)	5 (3.1 %)	0 (0 %)

[i] Includes late autoimmune diabetes in adults (LADA).
[ii] Diabetes-related admission causes include hyperglycaemia (N = 14), diabetic ketoacidosis (N = 7), diabetic foot infection (N = 4), hypoglycaemia (N = 2), new diabetes diagnosis (N = 2), and non-ketotic hyperosmolar coma (N = 2).
[iii] Not diabetes-related admission causes include cardiovascular diseases (N = 69), respiratory system diseases (N = 20), infections (N = 19), digestive system diseases (N = 15), genitourinary system diseases (N = 15), other emergencies (not included in the other categories) (N = 14), cancer (N = 12), endocrine diseases (N = 6), autoimmune diseases (N = 2), other (N = 5).
[iv] Measured from day of admission to November 14th, 2024 (i.e., the day of data collection); thus does not reflect the total duration of hospitalisation.
[v] Defined as Clinical Frailty Score ≥ 5, meaning "Living with mild frailty".
[vi] Diabetes complications were extracted from the responses to the Charlson comorbidity index (CCI) and included diabetes end-organ damage (N = 88),

myocardial infarction ($N = 49$), peripheral artery disease ($N = 31$), cerebrovascular disease ($N = 31$), and renal disease ($N = 87$).

[vii] Comorbidities were extracted from the responses to the CCI and included dementia ($N = 17$), COPD ($N = 29$), connective tissue disease ($N = 5$), peptic ulcer ($N = 13$), liver disease ($N = 58$), congestive heart failure ($N = 81$), hemiplegia ($N = 3$), cancer ($N = 39$), and AIDS ($N = 0$).

[viii] Technologies used before hospitalization include sensor-based continuous glucose monitoring (CGM) ($N = 4$) and hybrid closed-loop systems ($N = 1$); during the hospitalization: CGM only ($N = 3$), CGM in combination with an insulin pump ($N = 1$), and hybrid closed-loop systems ($N = 1$).

[ix] Obesity treatments used include GLP-1 receptor agonists ($N = 17$) and bariatric surgery ($N = 1$).

[x] Mean blood glucose levels showed a wide range from 64.7mg/dL to 451.3mg/dL.

($p>.05$). We did not observe any differences in terms of age, sex, glycaemic control, diabetes complications or multimorbidity. Rates of multimorbidity and/or frailty were generally high, affecting about 70 % of inpatients with type 2 diabetes with or without obesity. In both groups, about 65 % had CVD and/or renal disease and 40.7 % of people with type 2 diabetes generally and 25.0 % of those with diabesity had both. Considering this, we observed low rates of cardioprotective drug use prior to hospital admission: 39.8 % of inpatients with type 2 diabetes and CVD and/or renal disease were treated with a SGLT-2i and/or GLP-1ra before admission (5.3 % received a combination of both). Among people with diabesity and cardiovascular and/or renal disease 31.6 % received a SGLT-2i, 15.8 % a GLP-1ra, and one 5.3 % both (Fig. 1).

3.2. In-hospital diabetes management

Glycaemic monitoring and control. Most inpatients with type 2 (51.3 %) and type 1 diabetes (88.9 %) had a HbA1c level >6.5 % (48mmol/mol). However, this information was available for only two thirds of inpatients with type 2 diabetes and 90 % of inpatients with type 1 diabetes. POC blood glucose measurements were performed on average >6 times daily for inpatients with type 1 diabetes but <3 times daily for those with type 2. They were more consistently carried out before breakfast (85.0 %; T1D: 90.0 % vs T2D: 84.8 %) than before lunch or dinner (61.4 %; T1D: 80.0 % vs T2D: 62.7 %), or at night (24.6 %; T1D: 80.0 % vs T2D: 20.9 %). Mean blood glucose was 152.6mg/dL (8.47mmol/L) with significant variance between measurement time points. Average blood glucose levels were lowest at breakfast (138.7mg/dL / 7.7mmol/L), and highest at lunch (176.0mg/dL / 9.8mmol/L). The proportion of inpatients with blood glucose levels in the target range (100–180mg/dL / 5.6–10mmol/L) was 66.9 % (T1D: 44.4 % vs T2D: 69.1 %), but varied significantly, between the different measurement time points: 63.6 % at breakfast (T1D: 33.3 % vs T2D: 66.7 %), 61.4 % at dinner (T1D. 62.5 % vs T2D: 60.0 %), 57.5 % at lunch (T1D: 87.5 % vs T2D: 55.0 %) and 51.0 % at night-time (T1D: 50.0 % vs T2D: 51.4 %).

Pharmacological treatment. Most people with type 2 diabetes were treated with 1–2 glucose-lowering agents (Table 3). Half were treated with metformin before hospitalisation, about one third with SGLT-2i, and one quarter with DPP-4i. GLP-1ra were prescribed less, with around 12 % (Fig. 2). To examine treatment changes during hospitalisation we looked at the number of medications prescribed before versus during hospital stay (Table 3) and at newly prescribed and discontinued medications (Fig. 2). Slightly more than half of inpatients with type 2 diabetes received the same number of diabetes medications before and during hospitalisation. For about one quarter the number of medications either decreased or increased. Sulfonylureas were the most often discontinued agents (50.0 %; data not shown), followed by DPP-4i (40.2 %) and SGLT-2i (38.9 %). One third of inpatients with type 2 diabetes were started on insulin during hospitalisation and SGLT-2i, metformin, GLP-1ra, and DPP-4i were initiated for 2–11 %. Among inpatients with type 2 diabetes and cardiovascular and/or renal disease ($N = 113$) SGLT-2i were stopped for 39.4 % and initiated for 11.3 %, while GLP-1ra were discontinued or initiated for 12 % and 2.7 %, respectively (data not shown).

Table 2
Comparison of inpatients with type 2 diabetes and normal BMI, overweight, or obesity.

Variable	Inpatients with type 2 diabetes			Comparison A vs C: p-value (uncorrected)
	A – Normal BMI ($N = 49$)	B – Overweight ($N = 38$)	C – Diabesity ($N = 76$)	
Age	72.1 (13.4)	74.2 (9.4)	69.6 (12.5)	.287
Sex				.183
Female	19 (38.8 %)	15 (39.5 %)	40 (52.6 %)	
Male	30 (61.2 %)	23 (60.5 %)	36 (47.4 %)	
HbA1c (%) >6.5 % (>10mmol/mol)	7.7 (2.9) 11 (50.0 %)	7.4 (2.7) 15 (51.7 %)	7.0 (1.7) 28 (50.9 %)	.292
Admission cause[x]:				.003*
Diabetes-related	5 (10.2 %)	9 (23.7 %)	7 (9.2 %)	
Not diabetes-related	44 (89.8 %)	29 (76.3 %)	69 (90.8 %)	
Frail	28 (62.2 %)	24 (72.7 %)	55 (73.3 %)	.284
Cardiovascular complications	23 (47.9 %)	15 (41.7 %)	34 (44.7 %)	.872
Renal disease	23 (46.9 %)	15 (39.5 %)	35 (47.3 %)	1.0
Liver disease	10 (20.4 %)	11 (29.0 %)	27 (35.5 %)	.108
Home medication				
None	7 (14.6 %)	6 (15.8 %)	10 (13.2 %)	1
Metformin	20 (40.8 %)	11 (29.0 %)	46 (60.5 %)	.049*
DPP-4 inhibitor	12 (24.5 %)	14 (36.8 %)	25 (32.9 %)	.065
SGLT-2 inhibitor	2 (4.1 %)	12 (31.6 %)	13 (17.1 %)	.421
GLP-1 agonist	21 (42.9 %)	6 (15.8 %)	23 (30.3 %)	.057
Insulin		14 (36.8 %)		.212
Obesity treatment[x]	1 (2.1 %)	4 (10.8 %)	11 (14.5 %)	.031*
In-hospital therapy				
None	7 (14.6 %)	4 (10.5 %)	9 (12.0 %)	.888
Metformin	11 (22.5 %)	14 (36.8 %)	36 (47.4 %)	.009*
DPP-4 inhibitor	9 (18.4 %)	7 (18.4 %)	10 (13.2 %)	.591
SGLT-2 inhibitor	9 (18.4 %)	12 (31.6 %)	10 (13.2 %)	.153
GLP-1 agonist	3 (6.1 %)	6 (15.8 %)	24 (31.6 %)	.129
Insulin	30 (61.2 %)	18 (47.4 %)	36 (47.4 %)	.183
POC blood glucose measurements /day	3.0 (1.6)	2.5 (1.3)	2.4 (1.4)	.070
Blood glucose (mg/dL)	163.4 (60.6)	143.1 (33.3)	154.9 (58.1)	.406
Hypoglycaemia (<70mg/dL)	2 (4.9 %)	3 (8.6 %)	2 (2.7 %)	.617
Severe hypoglycaemia (<40mg/dL)	3 (7.1 %)	0 (0 %)	1 (1.4 %)	.142

Use of insulin. Most people with type 2 diabetes (66.1 %) were not treated with insulin before hospital admission. Insulin was initiated for 29.9 % of them ($N = 35$); with four receiving a long-acting insulin alone (11.4 %), twelve a short-acting insulin alone (34.3 %), and 17 (48.6 %) a combination of both. Compared to inpatients treated without insulin,

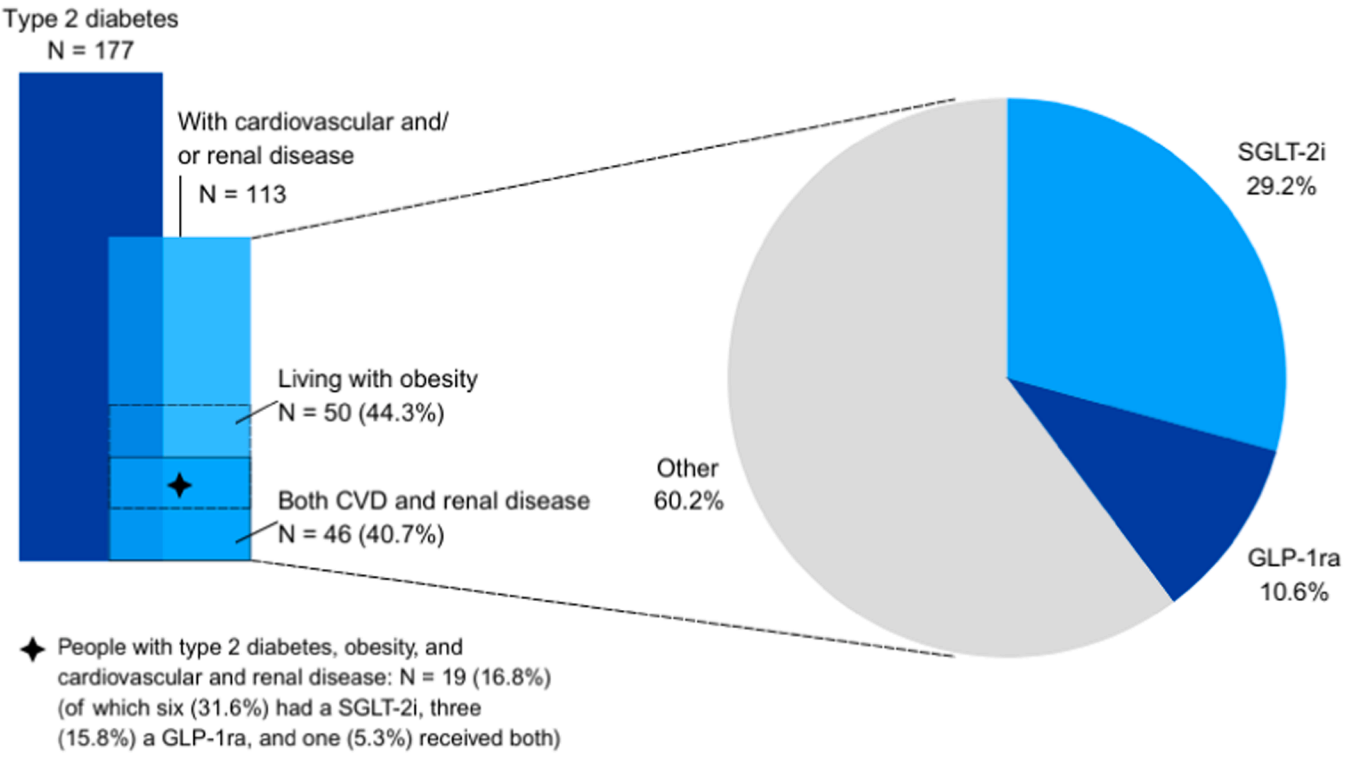


Fig. 1. Number of inpatients with type 2 diabetes and cardiovascular and/ or renal disease including of those who received treatment with a SGLT-2 inhibitor or GLP-1 receptor agonist.

Table 3
Changes in the number of diabetes medication used after hospitalisation.

No. medications before hospitalisation	Total	No. medications during hospital stay						No. people with more medications than before (escalation of treatment)
		0	1	2	3	4	5	
		24	46	64	27	13	3	
0	27	**14**	7	2	3	1	-	
1	45	6	**26**	12	1	-	-	
2	52	2	5	**36**	8	1	-	
3	28	2	6	7	**10**	3	-	
4	21	-	1	7	4	**8**	1	
5	4	-	1	-	1	-	**2**	**39**
No. people with less medications than before (de-escalation of treatment)							**42**	

they seemed to be younger (70.4 years vs 74.0 years), less often living with overweight (17.1 % vs 24.4 %) or obesity (40.0 % vs 47.6 %), and more often admitted due to diabetes-related problems (20.0 % vs 7.3 %). However, these differences were not statistically significant. We observed statistically significant differences in the frequency of POC blood glucose testing (2.9 times daily for inpatients with insulin vs 1.8 times ($p<.0001$)), and mean blood glucose levels (164.9mg/dL (9.2mmol/L) among inpatients with insulin vs 129.0mg/dL (7.2mmol/L) ($p<.01$)).

We included the following variables in the initial logistic regression model for predicting insulin treatment initiation: Age, sex, BMI, HbA1 level, admission cause, diabetes duration, creatinine level, albuminuria, frailty status, Charlson comorbidity index, in-hospital steroid use, and number of diabetes medications before hospitalisation. We iteratively removed the variable with the highest p-value until only statistically significant predictors remained ($\alpha=0.05$). In this process, the AIC of the model was reduced from 177.1 in the initial model to 168.1 in the final model, which included HbA1c, creatinine, and number of medications before hospitalisation. Higher HbA1c and creatinine levels increased the odds of initiating insulin, while a higher number of at-home medications lowered them (Table 4). *Re*-running the model with a subset of centres (with $N \geq 18$) yielded the same results (Supplemental Table 2).

Use of cardioprotective agents. Inpatients treated with cardioprotective agents during hospitalisation were more likely to have been treated with them before admission to hospital (SGLT2-i: 62.1 % vs 15.1 %, $p<.001$; GLP-1ra: 27.6 % vs 5.0 %, $p=.011$) and to be admitted due to CVD or diabetes-related reasons ($p=.011$). They also seemed to have a higher BMI (31.9 ± 6.9 kg/m2 vs 28.8 ± 6.3 kg/m2) and were more likely to take any glucose-lowering agent before hospitalisation (94.8 % vs 81.2 %); however, after correction for multiple testing these differences did not remain significant ($p>.05$).

We included the following variables in the initial logistic regression model for predicting use of SGLT-2i and/or GLP-1ra during hospitalisation: admission cause, age, sex, BMI, HbA1c, creatinine, CVD, renal disease, frailty status, and comorbidity index. The model identified cause of admission and BMI as significant predictors, with CVD-related admissions increasing the odds of prescribing SGLT-2i, diabetes-related admissions increasing the odds of prescribing a combination of SGLT-2i and GLP-1ra, and higher BMI increasing the odds of prescribing GLP1-ra alone or in combination with SGLT-2i (Table 5). When re-running the model with a subset of centres (with $N \geq 18$) the association of CVD-related admission and SGLT-2i use alone remained significant, indicating centre effects on medication use (Supplemental Table 3).



Fig. 2. In-hospital treatment changes for inpatients with type 2 diabetes (N = 177).
"Yes" and "No" indicate whether an agent was taken before admission to the hospital. The bold arrows indicate whether it was discontinued (pointing to the left to the stop sign) or continued (pointing downwards): The curved arrow indicates that the agent was initiated during hospital stay. "In-hospital" shows the number of inpantients receiving hte agent during hospital stay.

Table 4
Binomial logistic regression model predicting in-hospital initiation of insulin therapy.

	Odds ratio	95 % CI		p-value
		Lower limit	Upper limit	
HbA1c (%)	1.28	1.03	1.60	.035
Creatinine (mg/dL)	1.38	1.00	1.91	.057*
At home therapies (N)	0.68	0.49	0.95	.024

AIC of the initial model = 177.1; AIC of the final model = 168.1.
* Also removing Creatinine from the model led to an increase of AIC to 170.3.

4. Discussion

Our data enabled the characterisation of people with diabetes hospitalised in internal medicine wards in Europe on World Diabetes Day 2024 as well as the description of their diabetes-related hospital care. The findings reflect the increasing prevalence of diabetes and diabesity and provide insights into current (inpatient) type 2 diabetes management practices, particularly into the use of cardioprotective glucose-lowering agents and insulin and monitoring glycaemic status.

4.1. Cohort characteristics

About 30 % of inpatients in our study had diabetes, corresponding well with numbers reported over the past 20 years (18–40 %) (13,14,17–19). The proportion of people with diabetes among inpatients increased over time reflecting a general increase in diabetes prevalence (20). Overall, the composition of our sample is consistent with previously reported data: we observed a similar number of inpatients with diabetes complications (66 %), elevated HbA1c levels (52 %), obesity (42 %) or overweight (24 %), a similar distribution of men and women, and the same key causes of admission to hospital (cardiovascular, respiratory and infectious causes) (11,13,14,18,19,21).

The prevalence of obesity and overweight in our study is slightly higher than that among general inpatients (66 % vs 53 %) (1). However, the rate of overweight in our study was slightly smaller than among people with diabetes in general (24 % vs 27–38 %) (22,23) and we observed a higher proportion of inpatients with diabetes and underweight than is reported for the general population (3 % vs 1.6 %) (24). This may suggest that disproportionately fewer people with diabetes and overweight and more with underweight are being hospitalised, possibly emphasising the threat that underweight poses to people with diabetes and, particularly, to those with type 1. In our study, one third of inpatients with type 1 diabetes had underweight. Even though obesity rates are increasing among people with type 1 diabetes (25) no inpatient with type 1 diabetes had obesity in our study. This may be relevant to the "obesity paradox", a seemingly paradoxical, protective association between increased BMI and clinical outcomes in certain chronic conditions (26).

Among inpatients with type 2 diabetes, 50 % were living with diabesity, matching the general diabesity prevalence of 43–63 % (22,23). Although we did not find statistically significant differences between people with diabesity and those with type 2 and normal BMI, we observed a tendency among people with diabesity towards more females and people with frailty and liver disease, more frequent hospitalisations due to CVD or diabetes, higher use of metformin, SGLT-2i, or GLP-1ra, and more hypoglycaemia but fewer POC measurements during hospitalisation. These observations reflect general characteristics of people with diabetes and obesity, including a higher incidence among women (27) and increased risk of CVD and metabolic-associated fatty liver disease (5), and may indicate greater challenges in their inpatient management.

Table 5
Multinomial logistic regression model predicting in-hospital use of cardioprotective glucose-lowering agents.

In-hospital use of:	Cause of admission									BMI		
	CVD			Diabetes			Infection or emergency					
	Odds ratio	95 % CI	p-value	Odds ratio	95 % CI	p-value	Odds ratio	95 % CI	p-value	Odds ratio	95 % CI	p-value
SGLT-2i	6.01	2.05; 17.61	.001*	3.77	0.86; 16.58	.081	2.08	0.54; 8.03	.288	1.04	0.97; 1.12	.249
GLP-1ra	4.14	0.76; 22.44	.102	5.51	0.66; 45.92	.117	1.36	0.11; 17.09	.813	1.12	1.02; 1.24	.019*
Both	1.03	0.21; 5.10	.968	5.54	1.11; 27.71	.039*	0.69	0.07; 7.09	.754	1.13	1.03; 1.24	.014*

The reference levels for the categorical variables are "neither" for in-hospital use of cardioprotective treatment (i.e., no use of neither SGLT-2i nor GLP1ra during hospitalisation) and "other" for cause of admission (i.e., any other cause than CVD, diabetes-related, or related to an infection and/or an emergency).

* Statistically significant predictors of in-hospital use of cardioprotective glucose-lowering agents include CVD-related admission for use of SGLT2i, diabetes-related admission for use of a combination of SGLT-2i and GLP-1ra, and BMI for use of GLP-1ra alone or in combination with SGLT-2i. In a sensitivity analysis, only the association of CVD-related admission and in-hospital use of SGLT-2i remained significant (Supplemental Table 3), indicating centre-specific prescribing patterns.

4.2. Inpatient diabetes management: challenges and gaps

Our data indicate challenges with glycaemic control and monitoring: One third of inpatients with diabetes did not achieve the recommended blood glucose target range. Despite the clinical relevance of HbA1c upon admission and international guidelines recommending to use it as a basis for treatment and discharge decisions (4), this information was missing for 33 % of people with type 2 and 10 % of people with type 1 diabetes. Information on POC glucose levels was not available for 15–40 % of inpatients with diabetes, and for inpatients with type 2 diabetes, measurements were performed <3 times daily. This contrasts with guideline recommendations to perform POC glucose measurements ≥4 times daily and measure HbA1c in case HbA1c testing was not performed in the last three months (4). One explanation for lacking documentation of POC glucose measurements could be that some inpatients perform their own monitoring, without additional documentation by hospital staff. Guidelines encourage personal glucose monitoring (4), however, confirmatory POC tests are also recommended in this situation (4,28).

Our findings further indicate a substantial gap between the recommended and actual use of cardioprotective glucose-lowering therapies while showing that other guideline recommendations are being implemented. For example, metformin was continued for 34 % and initiated for 7 % of inpatients with type 2 diabetes, which is remarkable given that, previously, guidelines have recommended its discontinuation in the hospital setting (7,8). This observation confirms previous reports on the use of oral glucose-lowering agents, including metformin, in inpatient diabetes management (19). Emerging evidence on safety and benefits has led to recent guidelines no longer generally recommending pausing metformin during hospitalisation (4,11,29). Additionally, evidence on the effectiveness of SGLT-2i, DPP-4i, and GLP-1ra in the hospital setting has grown (4,9,10). Possibly reflective of this, we observed that 61 % of inpatients with prior use of SGLT-2i, 57 % of those with prior DPP-4i, and 75 % with GLP-1ra continued this treatment during hospitalisation, while 11 %, 2.3 %, and 25 % initiated treatment, respectively. We observed similar in-hospital use of insulin or DPP-4i and increased used of SGLT-2i and GLP-1ra as compared to previous surveys conducted in Spain (13,14), reflecting increasing availability and use of these drugs in recent years.

However, the number of inpatients with type 2 diabetes and cardiovascular and/or renal disease who received SGLT-2i and/or GLP-1ra in our study was alarmingly low, given that their use is strongly recommended for this group (6,30). Only 30 % were treated with SGLT-2i before and 25 % during hospitalisation, while only 10 % received GLP-1ra both before and during hospital stay. The proportion of inpatients receiving a combination of both agents was negligible (5 %), despite evidence suggesting superiority over monotherapy with regard to cardiac adverse events (31). It is unclear why many patients who would have benefitted from SGLT-2i and/or GLP-1ra therapy, including frail patients for whom SGLT-2i therapy has been shown to be particularly safe (32), or people with diabetes and COPD in whom SGLT-2i and GLP-1ra can reduce the risk of exacerbations (33), did not receive this treatment. Possible explanations may include the fact that SGLT-2i have been associated with an increased risk of diabetic ketoacidosis in acutely ill people (9) or a lag between clinical guidelines and their implementation, as reported by other studies (14). With regard to the limited use of GLP-1ra, this might reflect problems with access, their association with increased risk of aspiration during surgery and gastroscopy (34,35), and/or limited evidence for their in-hospital use (4). Another factor may be therapeutic inertia potentially due to uncertainty among clinicians regarding diabetes treatment adjustments (7) and the complexity of inpatient diabetes management, which is why guidelines recommend involving diabetes specialists (4).

Our analyses confirmed that higher HbA1c, creatinine, and number of medications used before hospitalisation led to treatment decisions in favour of insulin, despite the availability of newer therapies, which were more likely to be used in inpatients with high BMI and who were admitted due to CVD- or diabetes-related causes. These findings might indicate a certain degree of inertia in the implementation of new guidelines, which is valuable from a health systems and clinical practice perspective, as real-world data from hospital wards is rare. Insulin, which remains an important treatment option for people with diabetes experiencing hyperglycaemia on admission and/or during hospitalisation (4,8,11), was initiated for 30 % of inpatients with type 2 diabetes in this study. They had a higher HbA1c, continued to have higher blood glucose levels, and were monitored with POC glucose measurements more regularly than inpatients with type 2 diabetes without insulin. One third of inpatients with type 2 diabetes received SGLT-2i and/or GLP-1ra during hospitalisation. Individuals were more likely to receive these agents if they had already taken them before admission to hospital. These findings are partly consistent with known predictors of achieving glycaemic targets in hospital, including HbA1c level, type of diabetes treatment at home, and expected insulin resistance (4), for example in individuals with a higher BMI (5). While sensitivity analyses did not suggest a centre effect regarding predictors for initiating insulin therapy, they indicated that in-hospital use of SGLT-2i and GLP-1ra may be centre-/country-specific, potentially reflecting differences in availability and/or reimbursement policies. CVD-related causes for admission alone remained a predictor for the in-hospital use of SGLT-2i when fitting the regression model with a limited dataset.

4.3. Limitations

Several limitations warrant consideration: First, the study's cross-sectional study design meant that no causalities could be established. While the study identified aspects of inpatient diabetes management that appear particularly challenging to implement in real-world

practice, it cannot elucidate the underlying reasons for this.

Second, reliance on medical records and ward charts only may have introduced missing data, though the cause remains unclear. For example, measurements might not have been performed or not recorded. Particularly in the case of recent HbA1c readings, but also for other variables with incomplete data, the high rate of missing data is in itself a key finding: if recent HbA1c levels were unavailable to the research team, this information was likely also unknown/not of interest to treating clinicians and nurses, which indicates deficiencies in monitoring and/or documentation practices. This may further be related to the observed underuse of cardioprotective agents, as absent HbA1c data likely affected treatment modification/intensification.

Third, the sample is modest in size, mainly stems from Eastern and Middle European countries (e.g., Latvia, Austria, Czech Republic, and Germany), and lacked sociodemographic information due to data availability (e.g., in terms of ethnicity 76 % were White and 24 % were missing information, and information on diabetes training was only available for 15.5 %), which may have limited representativeness. However, the consistency of our findings with previously published national and international data supports their validity despite these limitations. The limited sample size further precluded detailed stratification (e.g., by cause of admission, nutrition modality, or insulin application methods, which might all have potentially influenced in-hospital treatment decisions) and may have reduced statistical power to detect significant differences between inpatients with type 2 diabetes and normal BMI versus obesity. Owing to this, we opted to compare people with diabesity to those with normal weight, assuming this would maximize potential differences and better reveal diabesity-specific characteristics. However, we acknowledge that this may be an oversimplification. Although we refrained from stratifying by centre/country due to a high variability in sample size ($N = 5 - 54$) we used sensitivity analysis to explore centre effects indicating that the use of SGLT-2i and GLP-1ra may be centre/country-specific. Considering this, contextual information about the different centres and healthcare systems would have been helpful for interpreting the results.

Despite these limitations, the study, which encompasses eight international hospitals, demonstrates the feasibility and successful collaboration within the EFIM network, particularly among the members of the new EFIM Diabetes and Obesity Working Group. This was the first pilot Diabesity Day Survey aimed to support methodological validation and refinement before scaling up in future survey rounds which we intent to conduct regularly and with more participating centres. Future analyses may benefit from study design adaptations: the relatively high amount of missing data, most likely due to our reliance on chart and medical record data, may be reduced by including a brief patient interview covering diabetes technology use, personal blood glucose monitoring, and last diabetes training, amongst others. Moreover, assessing person-reported outcomes such as psychological symptoms and quality of life, would add important information (5,36). BMI alone is not ideal for evaluating body composition and CVD risk as it does not accurately reflect the magnitude of adiposity (37,38). Future surveys should collect additional anthropometric and metabolic information, such as waist circumference or lipid profile (37). Lastly, future surveys should go beyond merely collecting data on patient monitoring and treatment and gather more comprehensive information, including hospitalisation duration, discharge and follow-up data as well as contextual factors such as institutional protocols and availability of endocrinologists and medications, including reimbursement policies, as this would enable a more nuanced examination of inpatient treatment practices and their impact on treatment outcomes.

5. Conclusion

Based on point-prevalence data of people with diabetes hospitalised during World Diabetes Day 2024 we characterise inpatients with diabetes and their management on European internal medicine wards. We showed that both obesity and underweight seem to add to people with diabetes' vulnerability for needing hospital treatment, emphasising the importance of supporting weight management for people with diabetes across the weight spectrum. Our data further unveiled limited use of SGLT-2i and GLP-1ra among at risk diabetes groups, low rates of POC glycaemic monitoring, and low availability of HbA1c readings. These findings indicate that clinical practice guidelines, in particular the recommendations on the use of cardioprotective glucose-lowering agents and glycaemic monitoring, are not sufficiently being implemented in everyday practice. Based on these gaps, we advocate for better clinical staff training and involvement of diabetologists on internal medicine wards. We urge hospital staff to be aware of the importance of knowing and monitoring the glycaemic status of inpatients with diabetes and to recognise the potential benefits of SGLT-2i and GLP-1ra to improve health outcomes for hospitalised people with diabetes.

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Data availability

The data set is available upon reasonable request from the corresponding author.

Declaration of competing interest

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejim.2025.106583.

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